

Tamás Bukits

Computer Vision Engineer & 3D Reconstruction Researcher

Donostia-San Sebastián, Spain | +36 30 410 5651 | tbukits@vicomtech.org

linkedin.com/in/bukitst | github.com/bukits | Portfolio: bukits.me

PROFILE

Industrial PhD Researcher at Vicomtech working at the intersection of computer vision, neural rendering, and extended reality, designing AI-driven pipelines that transform real-world people, objects, and environments into immersive digital experiences. My research integrates multimodal sensing, semantic scene understanding, volumetric capture, 3D Gaussian Splatting, and digital twins, with a focus on developing scalable, human-centred solutions for next-generation immersive applications.

PROFESSIONAL EXPERIENCE

Researcher, Interactive Media

Feb. 2024 – Present

Vicomtech

San Sebastián, Spain

- Progressed from Computer Vision Research Intern to Research Assistant and Industrial PhD Researcher in the Digital Media Department.
- Develop AI-driven pipelines for reconstructing real-world scenes, people, and objects into immersive XR environments.
- Research volumetric capture and 3D Gaussian Splatting, including methods that reduce dependence on Structure-from-Motion (SfM) and COLMAP-based initialization.
- Integrate semantic segmentation and multimodal scene understanding into live and offline volumetric media workflows.
- Contribute to European and industrial R&D projects in immersive media, digital twins, healthcare, and human-centred XR.
- Publish and present research at international conferences as both first author and co-author.

Software Engineer

Jan. 2022 – Jan. 2024

SwiconGroup Zrt.

Budapest, Hungary

- Built backend and frontend functionality for contract-management and HR workflow products using Node.js, TypeScript, Angular and MongoDB.
- Developed features for a Java-based sales management application and collaborated across product and engineering teams.

EDUCATION

Industrial PhD in Communication Technologies and Systems

Oct. 2025 – Present

Polytechnic University of Madrid

Madrid, Spain

Thesis: *Bringing Real World Entities Into Immersive Environments*: Designing AI-driven pipelines that digitise real-world people, objects, and environments using multimodal sensing, 3D scene understanding, and neural rendering to create scalable, high-fidelity immersive experiences for extended reality.

Erasmus Mundus Joint MSc in Image Processing and Computer Vision

Sep. 2022 – Jun. 2024

Pázmány Péter Catholic University · Autonomous University of Madrid · University of Bordeaux

Hungary · Spain · France

Grade: 4.11/5.00

Thesis: *AI Enhancement of Volumetric Captures* — a Gaussian Splatting pipeline designed to reduce reliance on Structure-from-Motion and COLMAP.

BSc in Computer Science Engineering

Sep. 2017 – Jun. 2021

Budapest University of Technology and Economics

Budapest, Hungary

Grade: 4.49/5.00

Thesis: *Algorithms for Smoothing Polyhedra* — a framework for generating smoothly connected surfaces from control polyhedra.

RESEARCH PROJECTS

BAZKARIA	Immersive telepresence for shared dining Human-centered XR research exploring remote dining, volumetric presence and interaction design in collaboration with Mugaritz. Role: Developed a real-time pipeline for capturing, processing, transmitting and rendering 3D holograms represented as point clouds, from depth-camera acquisition to display on HDMI-enabled devices.
SCENARIST	AI-powered synthetic data generation for safer public spaces End-to-end framework for creating photorealistic, automatically labelled synthetic datasets through 3D digitisation, Unreal Engine simulation and digital-twin technologies for AI training. Role: Contributed to 3D scanning, reconstruction and digitisation of real-world people and objects, enabling the generation of realistic digital assets for large-scale synthetic data creation using Gaussian Splatting
AMPLIFY	AI-generated immersive 3D memories Volumetric capture and evaluation of AI-based 3D reconstruction pipelines for immersive cultural experiences at the Gran Teatre del Liceu. Role: Developed and evaluated 3D reconstruction pipelines for objects and people using generative AI and 3D Gaussian Splatting.
INSPIRE	Volumetric digitization for XR training Volumetric capture and 3D reconstruction of people and environments to create realistic, reusable assets for immersive training applications. Role: Developed 3D reconstruction and volumetric digitization workflows for people and environments using 3D Gaussian Splatting.
AUTOPILOT	Digital surgical twins for robotic laparoscopy 3D reconstruction from laparoscopic video and digital-twin technologies supporting surgical planning, spatial understanding and increased robotic autonomy. Role: Contributed to laparoscopic 3D reconstruction and the development of digital surgical twins.
SYNERGIA	Adaptive virtual environments powered by generative AI Human-centered XR combining multimodal sensing, generative AI, human digital twins and intelligent agents to personalize training, rehabilitation and skills development. Role: Contributed to 3D reconstruction, human digital avatars and multimodal XR research.

PUBLICATIONS

A Sneak Peek into AI-Generated 3D Immersive Memories	ACM IMX 2026 · WiP
P. de Torres Coll, T. Bukits , J. Ruiz, I. Burguera Hidalgo, M. Zorrilla and S. Cabrero Barros.	
The XR Table: Envisioning the Future of Remote Dining Experiences Using Immersive Telepresence	ACM DIS 2025
A. Santos-Torres, P. de Torres Coll, T. Bukits , R. Perisé and S. Cabrero Barros.	
BAZKARIA: Eating Together Through Immersive Telepresence	EURO-XR 2025 · Poster
A. Santos-Torres, P. de Torres Coll, T. Bukits , A. Elosegui, I. Tamayo, R. Perisé, S. Cabrero Barros and M. Zorrilla.	
Beyond Bounding Boxes: 2D Semantic Segmentation for Live Volumetric Video Streaming	ACM Multimedia 2025
T. Bukits , A. Elosegi, A. Dominguez and S. Cabrero Barros.	

SKILLS

Research	3D reconstruction, 3D Gaussian Splatting, neural rendering, volumetric capture, semantic segmentation, computer vision, computer graphics, deep learning, digital twins, extended reality
Programming	Python, PyTorch, TensorFlow/Keras, C/C++, CUDA, OpenGL, TypeScript, Node.js, Java
Languages	Hungarian (native), English (full professional proficiency), Spanish (full professional proficiency), German (basic)

AWARDS & RECOGNITION

Jun. 2026	Selected alumni presenter at IPCVAI Day 2026	Bordeaux, France
2023 – 2024	Two Erasmus+ scholarships supporting academic exchange semesters	Madrid & Bordeaux
2022 – 2024	Image Processing and Computer Vision Excellence Scholarship - being selected in a cohort of 12 students from more than 1,000 applicants worldwide	Budapest, Hungary
2017 – 2021	Integrated MSc Scholarship - selected for a high-achieving cohort and maintained a 4.0/5.0 GPA	Budapest, Hungary